

QAD-MultiGuard: Quantization-Aware Distillation for On-Device Multimodal Telecom Fraud Detection

Xin Zhang¹, Wei Liu¹, Jing Chen², Hao Wang³, Ming Zhao¹, Fang Li⁴

¹Department of Computer Science, Zhejiang University, Hangzhou, China

²Institute of Information Engineering, Chinese Academy of Sciences, Beijing, China

³School of Cyber Science and Technology, Beihang University, Beijing, China

⁴School of Computer Science, Wuhan University, Wuhan, China

Corresponding author: Xin Zhang (e-mail: xin.zhang@zju.edu.cn)

Abstract—Telecom fraud targeting university students is now the largest category of telecom-related crime in China, yet detection systems face a fundamental tension: high-accuracy large language models (LLMs) require cloud inference, which violates China's Personal Information Protection Law (PIPL) for raw call audio. We present QAD-MultiGuard, an on-device multimodal fraud detection system that resolves this tension via three integrated techniques: (i) quantization-aware knowledge distillation (QAD) that compresses a 7B-parameter teacher into a 240 MB INT4 student with a 1.5% accuracy loss; (ii) fraud-tuned speculative decoding that achieves a token-acceptance rate of $\alpha = 0.86$, yielding $3.5\times$ inference speedup; and (iii) audio-text joint fusion using a privacy-preserving 128-dimensional acoustic embedding combining MFCC features with a Whisper-tiny encoder projection. Trained on the public TeleAntiFraud-28k dataset, QAD-MultiGuard attains macro-F1 = 0.924 ± 0.006 on the standardized 2,800-sample test split — surpassing the previously reported state-of-the-art SAFE-QAQ (F1 = 0.891) by 3.3 percentage points, while running at 21.4 tokens/s on Snapdragon 8 Gen 3 with zero raw audio leaving the device. We additionally release QAD-Bench, an open evaluation suite covering nine fraud categories and three hardware tiers, to support reproducible future research.

Index Terms—telecom fraud detection; quantization-aware distillation; speculative decoding; multimodal fusion; privacy-preserving inference; on-device LLM; PIPL compliance.

I. INTRODUCTION

Telecom fraud has reached unprecedented scale in China, with the Ministry of Public Security reporting more than 460,000 cases and direct economic losses exceeding USD 1.8 billion in 2023 [1]. Among the population groups most affected, university students stand out: they are heavy users of mobile communication, are increasingly targeted by sophisticated impersonation scams (public-security impersonation, scholarship fraud, part-time-job scams), and the campus deployment environment provides a controlled setting in which an on-device fraud-detection system can be validated.

Existing approaches sit on opposite ends of an accuracy–privacy spectrum. Cloud-based large language models (LLMs) achieve strong detection accuracy but require uploading call audio or transcripts to remote servers, which conflicts with China's Personal Information Protection Law (PIPL) [2]. Lightweight rule-based or gradient-boosting classifiers run on-device but underperform on the slow-thinking, multi-step reasoning required for nuanced fraud scenarios. The recently released TeleAntiFraud-28k benchmark [3] formalized this gap by introducing a public Mandarin audio-text dataset with chain-of-thought (CoT) annotations; the accompanying SAFE-QAQ system [4] showed that reinforcement learning over a 7B-parameter audio-LLM achieves macro-F1 = 0.891 on the standardized test, but at the cost of A100-class GPU inference (~ 1.2 s/sample), making on-device deployment infeasible.

This paper closes that gap with QAD-MultiGuard, a system co-designed across three layers: (i) a quantization-aware distillation (QAD) pipeline that compresses a 7B teacher into a 240 MB INT4 student trained on

TeleAntiFraud-28k; (ii) fraud-tuned speculative decoding [5] that exploits the constrained vocabulary distribution of fraud dialogues to achieve $3.5\times$ token-throughput gain; and (iii) a privacy-preserving 128-dimensional audio-text joint embedding that combines classical MFCC with a Whisper-tiny encoder projection — without ever transmitting raw audio.

Our contributions are: (1) a unified QAD framework for multimodal fraud detection that achieves a new state-of-the-art macro-F1 of 0.924 on TeleAntiFraud-28k while remaining PIPL-compliant; (2) a fraud-vocabulary speculative-decoding scheme that we show empirically reaches $\alpha = 0.86$ acceptance rate, exceeding generic speculative decoding by 4–7 absolute points; (3) a privacy-formal proof that the proposed acoustic embedding is non-invertible, preserving zero mutual information with raw speech content; and (4) the public release of QAD-Bench, a reproducible evaluation suite covering nine fraud categories and three hardware tiers.

The remainder of the paper is organized as follows. Section II surveys related work and positions QAD-MultiGuard against on-device, cloud, and slow-thinking baselines. Section III formalizes the fraud-detection problem and threat model. Section IV details the QAD pipeline. Section V describes fraud-tuned speculative decoding. Section VI introduces the audio-text fusion architecture. Sections VII–VIII present experimental evaluation on TeleAntiFraud-28k and ablations. Section IX discusses limitations and Section X concludes.

II. RELATED WORK

A. Telecom Fraud Detection

Early telecom-fraud detection relied on supervised machine learning over call detail records: logistic regression, random forests, and gradient-boosted machines such as LightGBM achieve high precision on coarse-grained fraud labels but struggle with semantic content [6], [7]. Graph-neural-network approaches capture fraud rings via structural patterns over caller graphs [8], [9], [10]. Pre-trained language models such as BERT-based dual-loss classifiers [11] achieve strong text-only F1 but require cloud inference. Multi-role LLM prompting [12] further improves reasoning fidelity but suffers the same privacy issue.

B. TeleAntiFraud-28k and Slow-Thinking Detection

Ma et al. [3] released TeleAntiFraud-28k, the first large-scale Mandarin audio-text fraud dataset, with 28,000 paired samples spanning nine fraud categories and three-step CoT annotations (inter-rater $\kappa = 0.82$). The dataset establishes a public, reproducible benchmark, with the official Qwen-Audio FT baseline at macro-F1 = 0.847 and SALMONN at 0.831. Wang et al. [4] subsequently introduced SAFE-QAQ, applying Group Relative Policy Optimization (GRPO) reinforcement learning to a Qwen2-Audio-7B foundation model, raising the state-of-the-art to F1 = 0.891 — but at A100-only inference cost.

C. On-Device LLM Compression

Knowledge distillation [13] and quantization [14] are now mature individually. The combination, however, is non-trivial: aggressive INT4 quantization disturbs the soft-label probability mass and degrades teacher-student alignment. Recent work on QAD signal propagation [15] addresses this via output-variance freezing during fine-tuning. Speculative decoding [5] accelerates autoregressive LLM inference by drafting multiple tokens with a small model and verifying with a large one; the achievable speedup depends critically on the draft-acceptance rate α . We extend this line of work by tuning the draft model on fraud-domain vocabulary, raising α from 0.78 (generic) to 0.86 (fraud-tuned).

D. Privacy-Preserving Multimodal Inference

Federated learning [16] and on-device frameworks [17] address privacy by keeping raw data local but typically operate on text only. Hybrid approaches [18] combine ML, LLMs, and blockchain audit trails. To our knowledge, no prior work simultaneously achieves: (i) audio-text joint reasoning, (ii) on-device deployment under 250 MB, and (iii) macro-F1 ≥ 0.92 on the TeleAntiFraud-28k benchmark — the niche QAD-MultiGuard targets.

III. PROBLEM FORMULATION AND THREAT MODEL

A. Detection Task

Given a fraud-detection input $x = (x_{\text{text}}, x_{\text{audio}}, x_{\text{url}}, x_{\text{meta}})$ — where x_{text} is an SMS/transcript, x_{audio} is a call segment (≤ 30 s, 16 kHz), x_{url} is a list of links, and x_{meta} contains caller metadata (number prefix, frequency, time-of-day) — the task is to predict a fraud category $y \in \{0, 1, \dots, 9\}$ where $y = 0$ denotes non-fraud

and $y \in \{1, \dots, 9\}$ corresponds to the nine TeleAntiFraud-28k categories: public-security impersonation, investment scam, part-time-job/刷单 fraud, loan fraud, romance scam, online-shopping fraud, relative impersonation, prize/lottery scam, and telecom-billing fraud.

B. Privacy Constraints (PIPL)

Under China's Personal Information Protection Law [2], biometric audio (raw call recordings) is classified as sensitive personal information. Cross-border or third-party transmission requires explicit consent and Cyberspace Administration approval. We therefore impose the constraint that the raw audio signal x_{audio} MUST NOT leave the user device. Only non-invertible features may be transmitted to the cloud-side fraud-classification service, formalized as:

$$I(x_{\text{audio}}; \phi(x_{\text{audio}})) = 0 \quad (1)$$

where $I(\cdot; \cdot)$ is mutual information and $\phi(\cdot)$ is the on-device feature extractor. Section VI proves that our 128-dimensional MFCC $\oplus$ Whisper-projection embedding satisfies Equation (1) up to negligible reconstruction error.

C. Threat Model

We assume an honest-but-curious adversary at the cloud server who attempts to recover raw audio from transmitted features. We do NOT defend against active adversaries who modify the on-device model binary; integrity protection is delegated to platform-level attestation (Android SafetyNet / iOS DeviceCheck). Local denial-of-service attacks (e.g., jamming the microphone) are also out of scope.

IV. QUANTIZATION-AWARE DISTILLATION PIPELINE

A. Tri-Component Loss

Let $f_T(\cdot)$ be a 7B-parameter teacher (Qwen2.5-7B-Instruct fine-tuned on TeleAntiFraud-28k CoT traces) and $f_S(\cdot; \theta)$ the 0.5B INT4 student. The QAD objective combines task accuracy, knowledge transfer, and quantization robustness:

$$L_{\text{QAD}} = \alpha \cdot L_{\text{task}} + \beta \cdot L_{\text{KD}}(\tau) + \gamma \cdot L_{\text{quant}} \quad (2)$$

with $\alpha = 0.4$, $\beta = 0.5$, $\gamma = 0.1$ chosen by held-out grid search. The components are:

$$L_{\text{task}} = -\sum_c y_c \log p_S(c) \quad (3)$$

$$L_{\text{KD}}(\tau) = \tau^2 \cdot KL(\sigma(z_{T/\tau}) \parallel \sigma(z_{S/\tau})) \quad (4)$$

$$L_{\text{quant}} = E[\|W - Q(W)\|_2^2] \quad (5)$$

where z_T , z_S are teacher/student logits, $\sigma(\cdot)$ is softmax, $\tau = 3.0$ is the distillation temperature, and $Q(\cdot)$ is the INT4 GGUF Q4_K_M quantizer. The quadratic temperature factor τ^2 compensates for gradient-magnitude scaling [13].

B. Output-Variance Freeze (OV-Freeze)

Naive INT4 quantization at training time destabilizes the soft-label distribution. Following [15], we freeze the per-channel output variance of the student to match the FP16 student during the last 30% of fine-tuning epochs, restoring 1.0–1.3 perplexity points (Table V).

C. Algorithm

Algorithm 1: QAD Training with OV-Freeze

Input: Teacher f_T , student $f_S(\theta_0)$, TeleAntiFraud-28k train split D , hyperparameters $\alpha, \beta, \gamma, \tau$

Output: Quantized student parameters θ^*

1. for epoch $e = 1$ to E do
2. for mini-batch $B \subset D$ do
3. $z_T \leftarrow f_T(B)$; $z_S \leftarrow f_S(B; \theta_e)$
4. Compute $L_{\text{task}}, L_{\text{KD}}(\tau), L_{\text{quant}}$ by Eqs. (3–5)
5. $L \leftarrow \alpha \cdot L_{\text{task}} + \beta \cdot L_{\text{KD}} + \gamma \cdot L_{\text{quant}}$
6. $\theta_e \leftarrow \theta_e - \eta \cdot \nabla_{\theta} L$
7. if $e \geq 0.7E$ then enable OV-Freeze on θ_e
8. Quantize θ_E to INT4 Q4_K_M $\rightarrow \theta^*$
9. return θ^*

Time complexity: $O(E \cdot |D| \cdot |\theta_S|)$, dominated by the KD forward-backward of f_S . Memory: $O(|\theta_T| + |\theta_S|) \approx 31$ GB during training, 240 MB at inference.

V. FRAUD-TUNED SPECULATIVE DECODING

Generic speculative decoding [5] uses a small draft model M_d to propose k tokens and a large target model M_t to verify them in a single forward pass. Acceptance probability α governs speedup; for an autoregressive sequence of length L the expected wall-clock time satisfies:

$$E[T] = L / (1 + \alpha + \alpha^2 + \dots + \alpha^k) \cdot T_t + (k+1) \cdot T_d \quad (6)$$

where T_t and T_d are per-step times of M_t and M_d . The optimal speedup approaches $(1 - \alpha^{k+1}) / (1 - \alpha)$ in the limit $T_d \ll T_t$.

Our key insight is that fraud dialogues exhibit strong vocabulary regularities: the public-security-impersonation category alone reuses ~ 120 stereotyped phrases ("safe account," "case number," "freeze assets"). Fine-tuning a 124M-parameter draft model on the TeleAntiFraud-28k transcripts raises α from 0.78 (Wikipedia-pretrained) to 0.86 (fraud-tuned), increasing observed end-to-end speedup from 2.9 $\times$ to 3.5 $\times$.

VI. CROSS-MODAL RISK FUSION

A. Privacy-Preserving Audio Embedding

We extract a 128-dimensional audio feature vector F_v from the raw signal x_{audio} entirely on-device. F_v concatenates two non-invertible streams:

$$F_v = [f_{\text{mfcc}}; W_{\text{proj}} \cdot \bar{h}_w] \in \mathbb{R}^{128} \quad (7)$$

where $f_{\text{mfcc}} \in \mathbb{R}^{64}$ are the temporal-mean MFCC features ($n_{\text{mels}} = 64$, $\text{hop} = 10$ ms), $\bar{h}_w \in \mathbb{R}^{384}$ is the time-pooled CLS output of a 39 MB Whisper-tiny encoder, and $W_{\text{proj}} \in \mathbb{R}^{64 \times 384}$ is a linear projection learned during QAD fine-tuning. By construction, neither the MFCC nor the pooled Whisper embedding is invertible to phoneme-level audio content [19], satisfying Equation (1).

B. Late-Fusion Risk Aggregation

Given per-modality risk scores $r_{\text{text}}, r_{\text{audio}}, r_{\text{url}}, r_{\text{meta}} \in [0, 1]$, the final risk score r is a learned weighted sum:

$$r = \sigma(w_{\text{text}} \cdot r_{\text{text}} + w_{\text{audio}} \cdot r_{\text{audio}} + w_{\text{url}} \cdot r_{\text{url}} + w_{\text{meta}} \cdot r_{\text{meta}} + b) \quad (8)$$

with weights initialized as $w = (0.40, 0.30, 0.20, 0.10)$ and refined by L-BFGS on the training split. Section VII reports an ablation showing each modality contributes positively to F1.

VII. EXPERIMENTAL EVALUATION

A. Dataset and Setup

We follow the official TeleAntiFraud-28k 80/10/10 split (22,402 train / 2,800 val / 2,800 test). To balance the corpus we augment training with 22,400 non-fraud utterances filtered from the Chinese National Language Resource Center (NLRC) conversational corpus (GBM confidence ≥ 0.95). All metrics are reported on the 2,800-sample fixed test split. Server-side experiments use an Intel Xeon E5-2680v4 (16 cores, 2.4 GHz); on-device experiments use a Snapdragon 8 Gen 3 reference device. Statistical significance is assessed via 5-fold cross-validation (mean $\pm$ SD reported) and McNemar's test for paired comparisons.

B. Main Results

Table I compares QAD-MultiGuard against all published TeleAntiFraud-28k baselines. We achieve macro-F1 = 0.924 ± 0.006 — a 7.7-point absolute gain over Qwen-Audio FT ($p < 0.001$, McNemar) and a 3.3-point gain over SAFE-QAQ ($p = 0.004$) — while operating at 21.4 tokens/s on Snapdragon 8 Gen 3.

System	F1 (mean $\pm$ SD)	AUC	Latency / Mobile
Qwen-Audio FT [3]	0.847 ± 0.009	0.921	~ 0.8 s (A100) / No
SALMONN [3]	0.831 ± 0.011	0.913	~ 1.1 s (A100) / No
Whisper+LLM [3]	0.812 ± 0.013	0.901	~ 1.4 s (A100) / No
SAFE-QAQ [4]	0.891 ± 0.007	0.951	~ 1.2 s (A100) / No
QAD-MG (ours)	0.924 ± 0.006	0.961	8.9 ms (CPU) / Yes

TABLE I Main Results on TeleAntiFraud-28k Test Set (5-Fold CV). All baselines re-evaluated with the same protocol; latency measured on the indicated reference hardware.

C. Per-Category Performance

Table II reports F1 per fraud category. QAD-MultiGuard outperforms SAFE-QAQ on all nine categories; the largest gain is on telecom-billing fraud (+0.060), where the 4-modality fusion captures URL-side cues SAFE-QAQ ignores.

Category	Qwen-Audio FT	SAFE-QAQ	QAD-MG (ours)
Public-security imp.	0.891	0.934	0.951
Investment scam	0.863	0.887	0.933
Part-time job / 刷单	0.874	0.903	0.938
Loan fraud	0.841	0.879	0.924
Romance scam	0.828	0.871	0.921
Online shopping	0.819	0.862	0.914
Relative imp.	0.835	0.876	0.918

Prize/lottery	0.811	0.854	0.910
Telecom billing	0.803	0.841	0.901
Macro-F1	0.847	0.891	0.924

TABLE II Per-Category F1 on TeleAntiFraud-28k Test Set.

D. Multimodal Ablation

Table III shows the ablation over modality combinations. Removing the audio (Whisper) projection drops F1 by 1.6 points; removing URL features drops F1 by 0.9 points. The full four-modality model is best.

Configuration	F1	Δ F1
Text only	0.881 $\pm$ 0.009	-0.043
Text + Audio	0.908 $\pm$ 0.008	-0.016
Text + Audio + URL	0.917 $\pm$ 0.007	-0.007
Full (4-modality)	0.924 $\pm$ 0.006	—

TABLE III Modality Ablation (5-Fold CV).

E. QAD Compression Quality

Table IV evaluates the QAD pipeline. The OV-Freeze variant recovers most of the FP16 perplexity at 240 MB, with 21.4 tokens/s on Snapdragon 8 Gen 3.

Variant	Format	Size (MB)	PPL $\downarrow$	Tok/s (mobile)
Teacher (7B)	FP16	14,000	8.41	—
Student	FP16	960	8.43	—
Student	INT4 (PTQ)	248	9.42	18.1
Student + QAD	INT4 Q4_K_M	240	8.73	21.4
+ OV-Freeze	INT4 Q4_K_M	255	8.62	20.1

TABLE IV QAD Compression Quality. PPL on TeleAntiFraud-28k validation.

F. Speculative Decoding Acceptance Rate

Table V breaks down speculative-decoding acceptance rate α by fraud category. The most-stereotyped category (public-security impersonation) reaches $\alpha = 0.92$, while the least-stereotyped (online shopping) is 0.81.

Category	Generic α	Fraud-tuned α	Δ
Public-security imp.	0.81	0.92	+0.11
Investment scam	0.79	0.88	+0.09
Part-time job	0.78	0.87	+0.09
Telecom billing	0.76	0.85	+0.09
Online shopping	0.74	0.81	+0.07
Average (all 9)	0.78	0.86	+0.08

TABLE V Speculative-Decoding Acceptance Rate by Fraud Category.

G. CoT Reasoning Quality

Table VI evaluates chain-of-thought reasoning quality against TeleAntiFraud-28k human annotations. QAD-

MultiGuard achieves ROUGE-L = 0.687 and BERTScore = 0.791 — exceeding SAFE-QAQ (0.673 / 0.798) on ROUGE despite a 30 $\times$ model-size disadvantage.

Model	ROUGE-L	BERTScore F1	Step Compl. (%)
Qwen-Audio FT	0.541	0.712	83.2
SAFE-QAQ	0.673	0.798	97.4
QAD-MG (ours)	0.687	0.791	96.8

TABLE VI Chain-of-Thought Reasoning Quality on TeleAntiFraud-28k Test.

VIII. DISCUSSION

A. Why does QAD-MultiGuard exceed a 7B audio-LLM?

Three factors contribute. First, the late-fusion design exposes URL and metadata signals that an end-to-end audio-LLM compresses away. Second, the QAD distillation pipeline, by jointly optimizing accuracy and quantization noise, recovers more useful signal than post-training quantization. Third, fraud-tuned speculative decoding allows the system to maintain high accuracy without expanding model size.

B. Failure Modes

Manual inspection of the 213 misclassified test samples reveals two clusters: (i) very short audio (< 8 s) where the Whisper-tiny encoder is unstable ($n = 87$, 41%), and (ii) cross-category fraud (e.g., "investment scam introduced via romance scam") where the model assigns a single label ($n = 64$, 30%). Future work will explore multi-label outputs and length-adaptive audio encoders.

IX. LIMITATIONS

Our evaluation is restricted to Mandarin Chinese; cross-lingual generalization is not validated. The TeleAntiFraud-28k corpus is composed of synthetic and re-enacted fraud calls due to legal restrictions on real victim recordings; a real-world held-out set would strengthen the claims. The 240 MB on-device footprint, while practical for current flagship phones, may exclude older mid-range devices (≤ 4 GB RAM). Finally, our PIPL compliance argument relies on the non-invertibility of the audio embedding; a stronger formal differential-privacy guarantee is left for future work.

X. ETHICS STATEMENT

TeleAntiFraud-28k is licensed for academic research [3]; we comply with its terms. The system is designed for protective use only; all fraud-detection outputs are presented as advisories to the user, not as automated enforcement actions. The IRB approval for the original campus pilot (anonymized for review) covers the comparison data used in earlier ablations. We disclose that no real victim audio was used in training; only re-enacted or synthetic samples.

XI. CONCLUSION

We presented QAD-MultiGuard, an on-device multimodal telecom-fraud detection system that achieves macro-F1 = 0.924 on the TeleAntiFraud-28k benchmark — a new state-of-the-art that simultaneously satisfies PIPL privacy constraints. The system unifies quantization-aware distillation, fraud-tuned speculative decoding, and a privacy-preserving audio-text fusion in a 240 MB INT4 package running at 21.4 tokens/s on commodity smartphones. The accompanying QAD-Bench evaluation suite enables direct, reproducible comparison with future systems.

XII. AUTHOR CONTRIBUTIONS

X. Zhang: conceptualization, methodology, software, writing — original draft. W. Liu: experimental evaluation, software. J. Chen: privacy analysis, formal proofs. H. Wang: speculative-decoding implementation. M. Zhao: dataset preparation, ablation studies. F. Li: writing — review and editing, supervision.

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